# Hybrid Computing Resources and Environments

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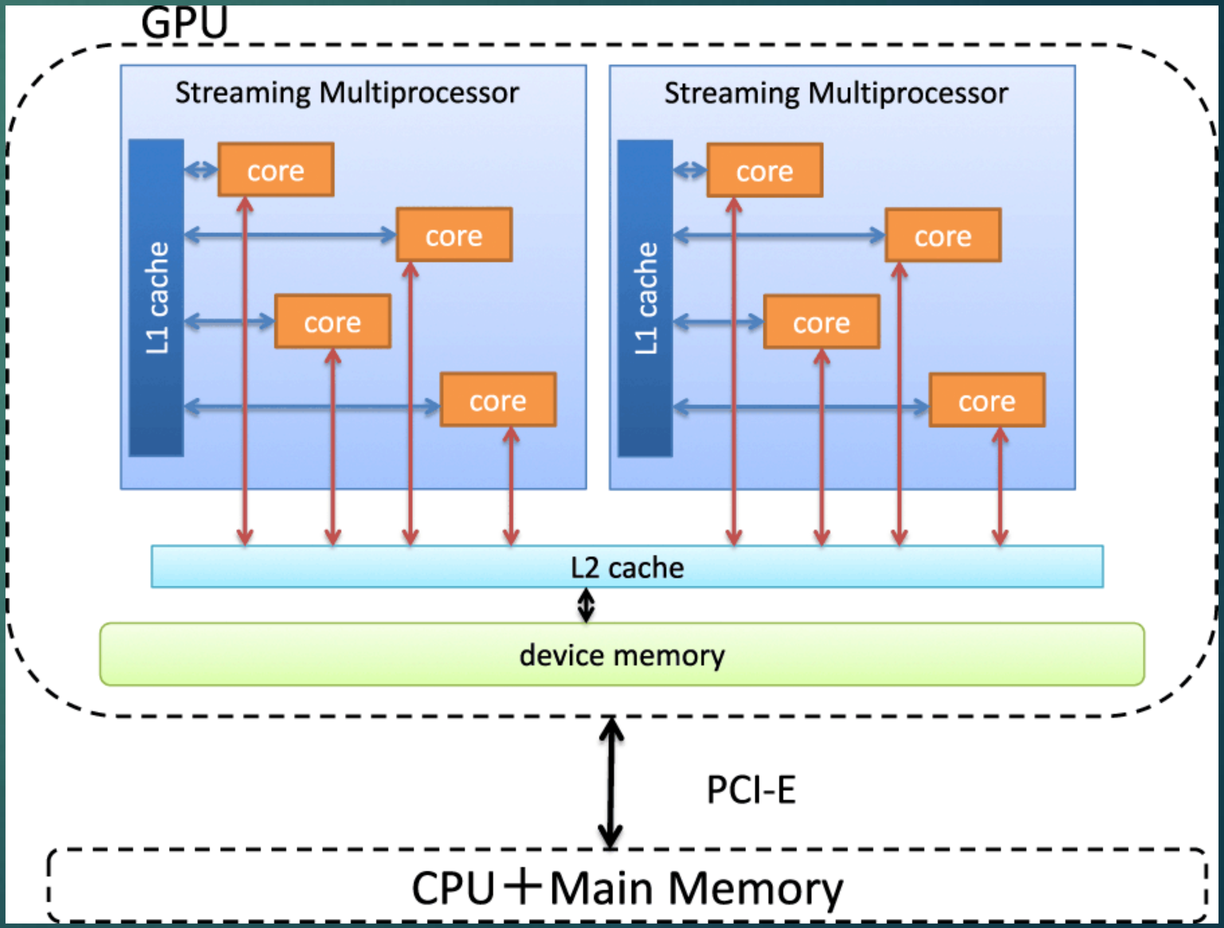
# Topics

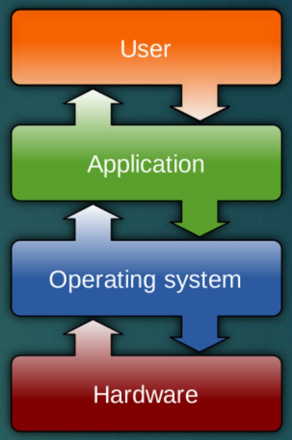


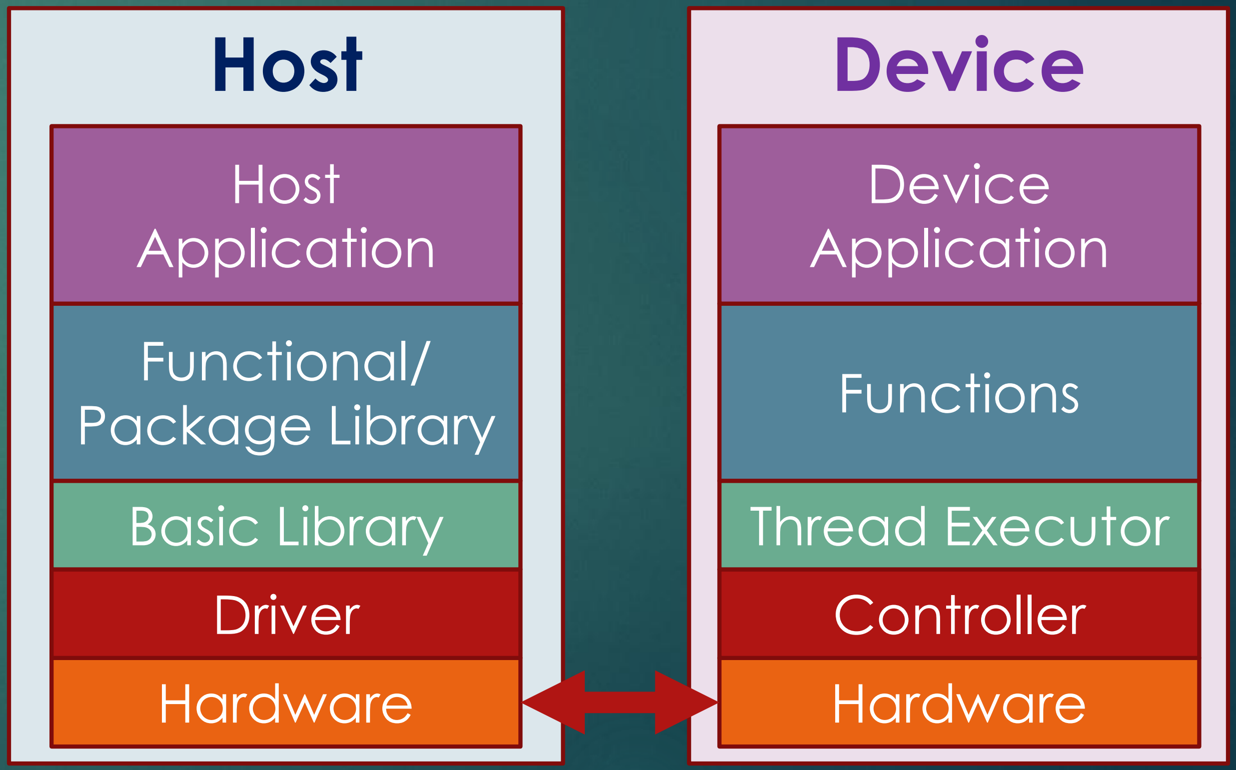
- Computing Resources and Environments
- Hardware Resources
- Software Resources
- Physical Environments
- Resource Calculations
- Conclusion



# Computing Resources and Environments

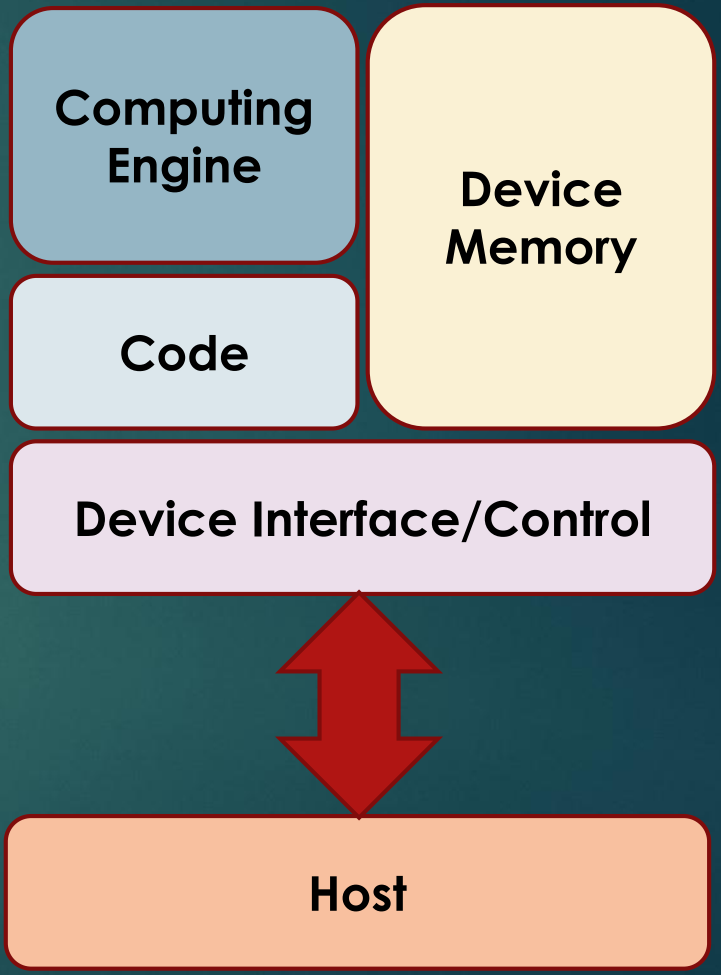
- Hardware
- Software
- Data
- Environment
- People
- Operation/Processes



# Computing Hardware Resources

- Host
  - CPU
  - Main Memory
  - Storage – SSD/HDD
  - Communication Network – Hosts connected
- Devices
  - Compute Engines – Size / Speed
  - Memory – DRAM Size
  - Communication – Transfer Channel Speed

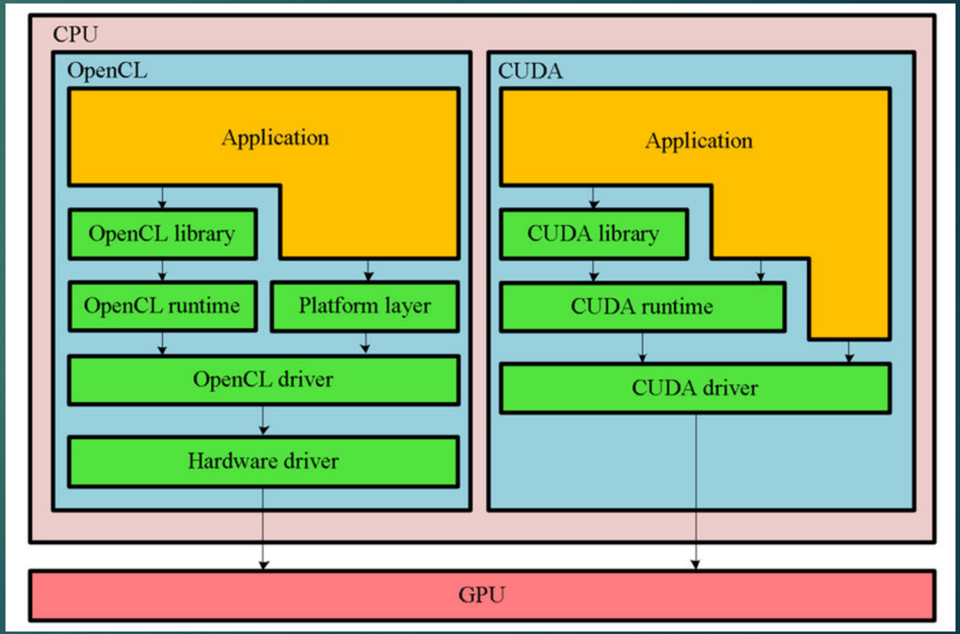


# Computing Software Resources



- Operating System (OS)
- Framework and Development Tools
- Drivers and Libraries
- Debugging Tools (Option)
- Applications (Source Codes)

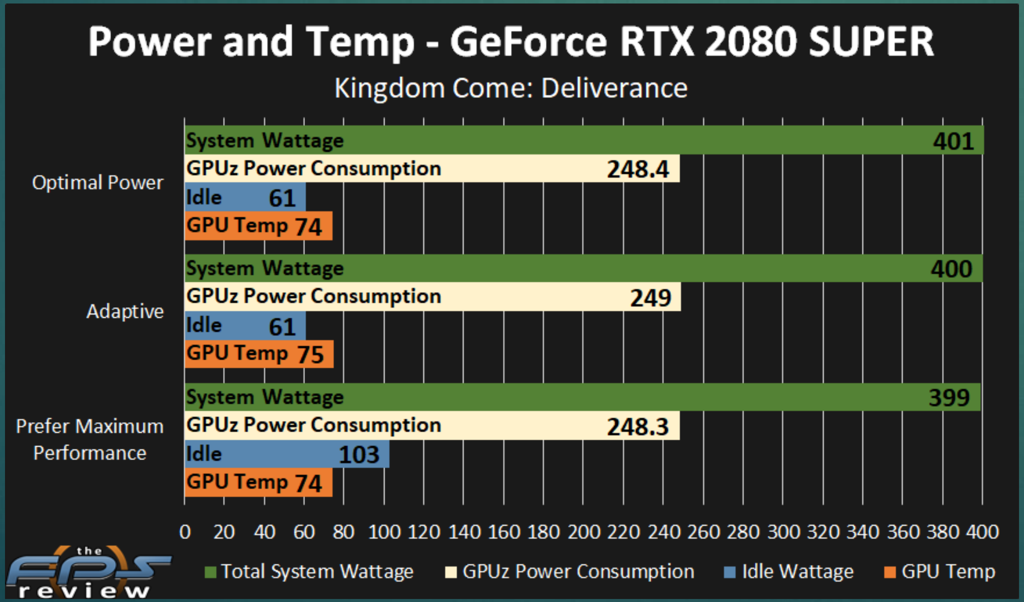
- **Depend on Computing Devices**



# Physical Environments



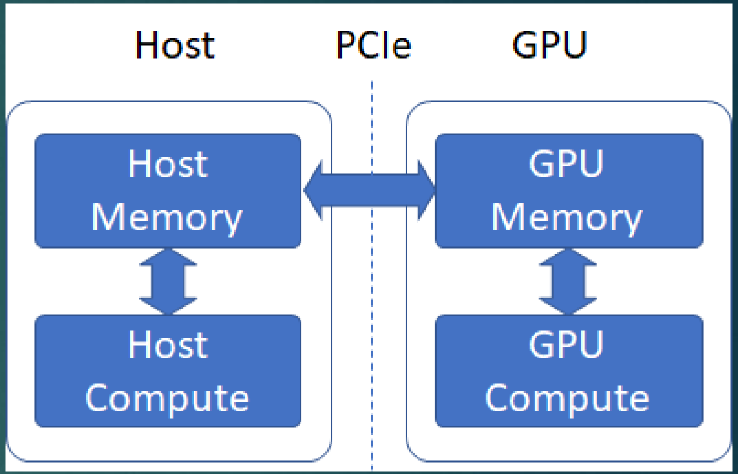
- Power – Electrical Power
  - All components – Compute Engines
- Cooling – Heat ventilation
  - All components – Compute Engines

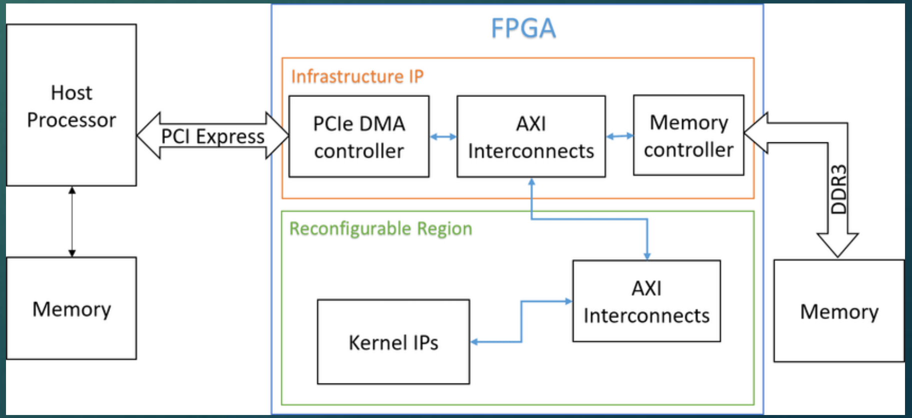



# Calculations



- Device Resources
  - Compute Speed – FLOPS (GFLOPS, TFLOPS)
    - Each Device Throughput x Number of Devices
    - Enough for computing all data in-time (Required Time)
  - Device Memory – Size and Bandwidth
    - Store partial data
    - Memory Bandwidth – Fast enough for computing speed
  - Host-Device Transfer Speed
    - PCI Express
    - Other high speed connection

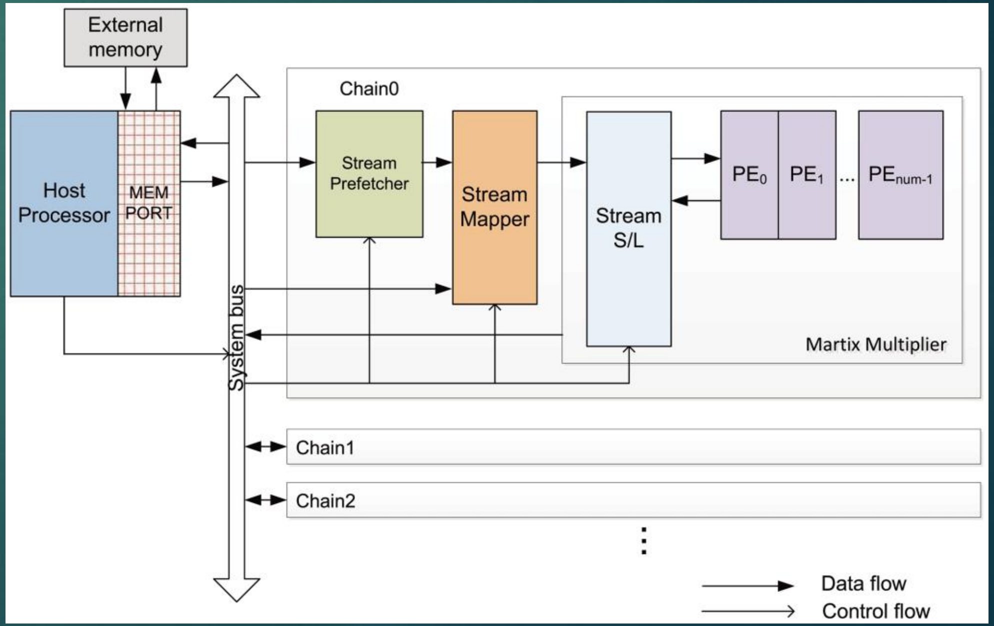


# Calculations



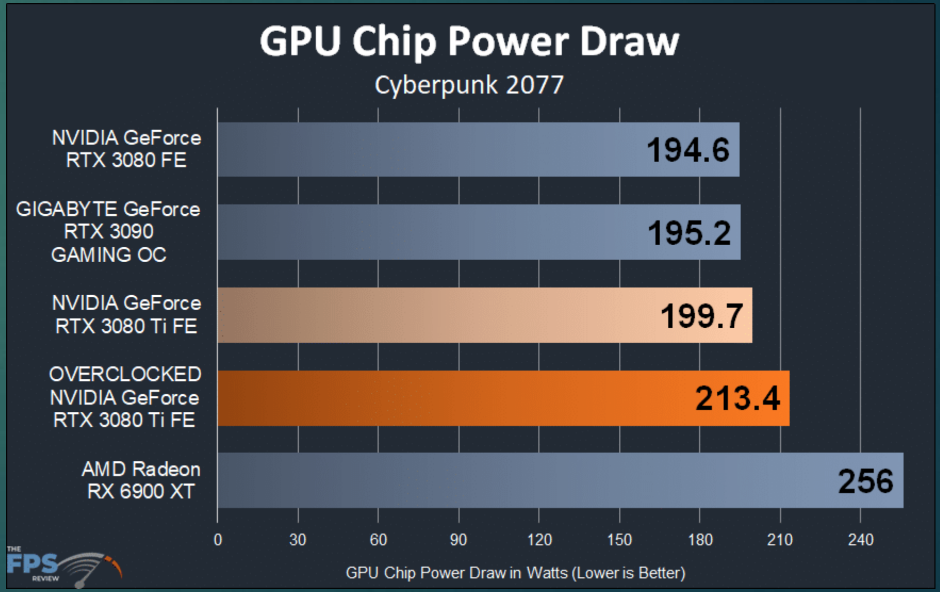
- Host Resources
  - Host Processor – CPU Speed
    - Fast Enough to control and distribute compute and data workload
  - Host Memory – RAM
    - Temporary Store for Data – Transferring to Devices
    - Size and Speed
  - Host Storage – SSD/HDD
    - Store all Data – Input and Results
    - Size and Speed
  - Hosts Interconnections
    - Multiple Hosts Communication – Scalable



# Calculations



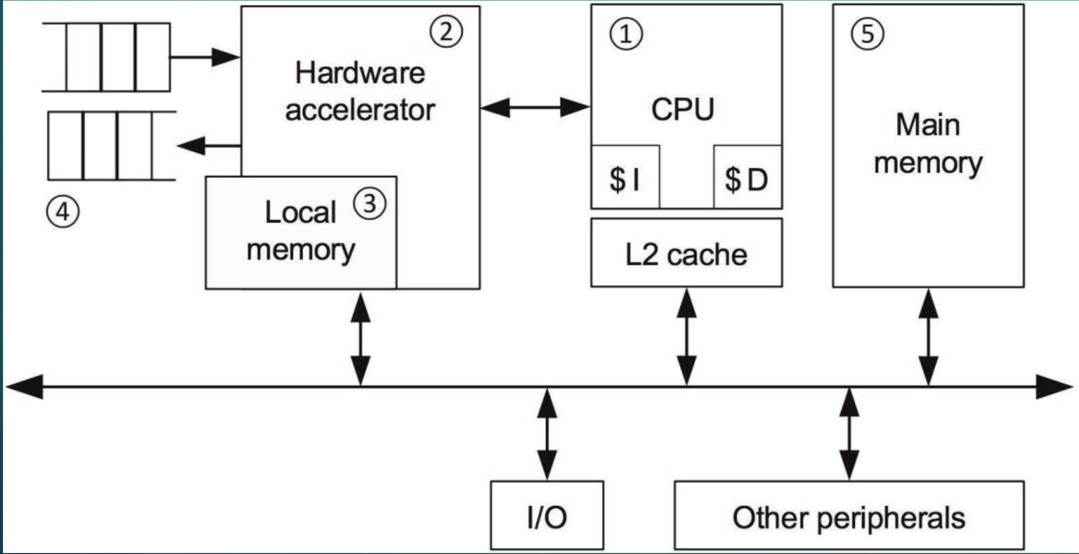
- Physical Environments
  - Total Electrical Power – Kilo/Mega-Watts
  - Sum of Power for Each Device + Host Power
  - Additional for Peak Load – Spare Power => 10%-20%
- Cooling
  - Dissipating Heat from Engines to outside world
  - Air Cooling vs Liquid Cooling
  - Heat Transfer - BTU

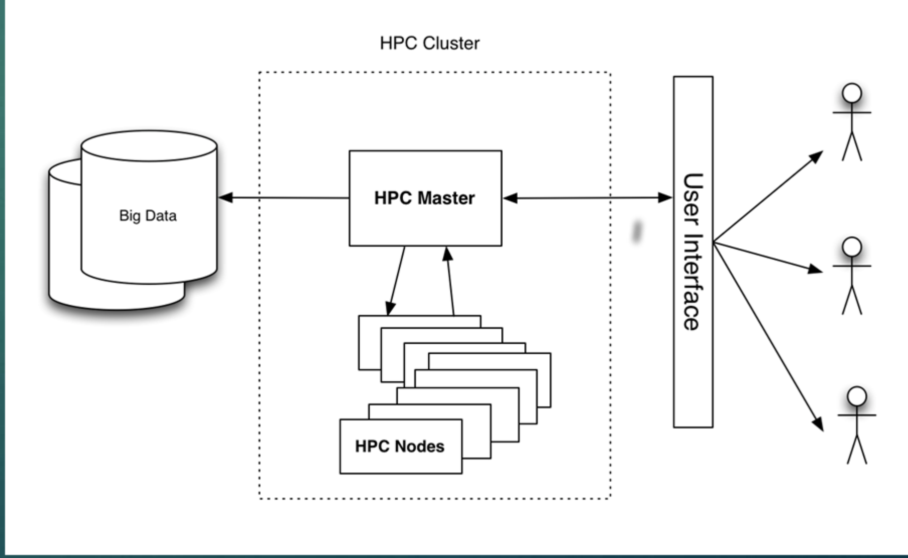


# Conclusion



- Prepare Resources and Environments for Computing
- Hardware – Devices and Hosts
- Software – Depend on Devices
- Physical Environments – Power and Cooling
- Calculate Sizes of Resources and Environments



# END

QUESTIONS & ANSWERS